SAHAKAR-LINK

Product Requirements Document (PRD)

Hardware-agnostic, offline-first cooperative training ecosystem

Problem Statement
SIH26087 — AI & LMS-Enabled Cooperative Capacity Building, ERP & Employment Ecosystem

Owner / Department
Ministry of Cooperation / National Council for Cooperative Training (NCCT)

Design status
Hackathon-ready reference architecture; NCCT-specific unknowns are explicitly marked as assumptions or open questions.

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1. Executive Summary

Sahakar-Link is a hardware-agnostic digital operating system for cooperative training centers. It unifies programme registration, nominations, trainee profiles, multilingual learning, attendance, assessments, certification, skill evidence, employer workflows and employment outcomes while preserving learning continuity during intermittent connectivity.

The design deliberately separates governance from frontline operations: centralized cloud services provide system-wide control and analytics, while a software Local Fabric runs on available center hardware to cache learning content, persist operational data and queue changes for later synchronization. This makes the product resilient without requiring proprietary hardware.

Architecture decision

"Hardware-agnostic" is a product/architecture decision, not a claim that the SIH Hardware category is automatically satisfied. Because the official PS is categorized Hardware and states Software + Hardware, the team should confirm the category interpretation with the SIH SPOC before final submission. The platform remains capable of using optional low-cost local infrastructure without depending on a specific appliance.

2. Problem & Opportunity

The official SIH problem statement describes fragmented/manual training operations, inadequate trainee tracking, limited learning analytics, poor employment linkage and inefficient programme administration. It asks for a centralized, scalable, web-based ecosystem covering training management, e-learning, analytics, skill development, digital literacy and employment exchange services. [S1]

2.1 Core user problem

Today the platform must solve one connected lifecycle rather than a set of isolated screens:

Nomination → Training → Learning → Attendance → Assessment → Certification → Skill Evidence → Employment → Outcome Tracking

2.2 Product opportunity

Research supplied with the project supports offline-capable learning, local caching, queued writes and incremental synchronization for constrained environments. It also identifies synchronization integrity as a major operational risk and favors simple append-only change tracking plus server reconciliation for a hackathon MVP. [S2]

3. Product Vision and Principles

Vision: "Make every cooperative training center digitally operable, even when connectivity is unreliable, and make every training outcome traceable from nomination to employment."

Principle	Meaning
Edge-first	Frontline learning and operational capture must continue when the network is unavailable.

Principle	Meaning
Event-driven	Important changes are captured as durable, replayable events.
Evidence-based	Skills, credentials and outcomes are linked to underlying evidence.
Hardware-agnostic	The software runs across laptops, desktops, tablets and smartphones; a local node is optional.
Human-in-the-loop	AI assists; people remain responsible for high-impact decisions.
Integration-ready	External NCCT/government systems are connected through adapters, not hard-coded assumptions.
Outcome-oriented	Analytics connect training activity to certification and employment outcomes.
Graceful degradation	When one dependency fails, the core training workflow remains usable.

4. Users and Roles

Role	Primary responsibilities
NCCT Admin	System-wide oversight, programme governance, analytics, policy/configuration
Regional / Institution Admin	Programme setup, nominations, batches, timetable, hostel, logistics, staff
Trainer	Classes, attendance, assessment, learner progress, alerts
Trainee	Learning, attendance, assessment, credentials, jobs, career support
Employer / Recruiter	Employer verification, job posting, candidate discovery, applications, outcomes
Auditor	Read-only access to audit events, credential history and operational evidence
Platform Operator	Technical operations, device/node health, monitoring and support

5. Goals / Non-Goals

5.1 Goals

- Cover every major expected feature in SIH26087 through a coherent lifecycle.
- Keep learning usable during network outages.
- Provide a trustworthy attendance path using face/liveness or QR with manual exception handling.
- Produce verifiable credentials and skill evidence without blockchain dependency.
- Give employers a simple, explainable route to discover qualified candidates.

- Measure outcomes beyond training completion.
- Remain deployable on existing devices and integration-ready for future government systems.

5.2 Non-goals for MVP

- Replacing all existing national education/credential platforms.
- Aadhaar identity integration without an explicit authorization/requirement.
- A nationwide proprietary hardware rollout.
- CRDT-everywhere or multi-master ERP semantics.
- Autonomous hiring decisions by AI.
- Blockchain-based credential infrastructure.

6. Solution Overview

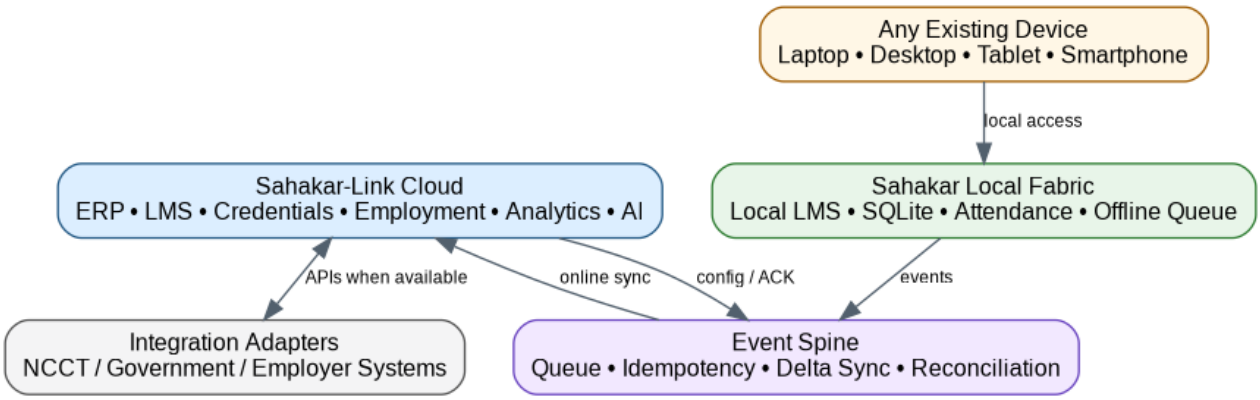


Figure 1. Hardware-agnostic cloud + Local Fabric architecture.

6.1 Product surfaces

Surface	Primary users	Core value
NCCT Control Center	NCCT admins	Nationwide programme and outcome visibility
Institution Console	Center admins	Daily programme operations
Trainer Workspace	Trainers	Teaching, attendance and assessment
Trainee App/PWA	Trainees	Learning, progress, skills and employment
Recruiter Console	Employers	Verified candidate discovery and hiring workflow
Local Fabric	Training center	Offline learning, local persistence and sync
Verification Portal	Employers/auditors	Credential verification and revocation checks

7. Module Requirements

ID	Module	Requirement
PRD-FR-001	Programme & nomination	Create programmes, define eligibility, open/close nomination window, accept nominated participants, track nomination status.
PRD-FR-002	Profile management	Maintain institution, trainee, trainer and employer profiles with RBAC and versioned updates.
PRD-FR-003	Batch management	Create batches, assign programme/trainees/trainers and lifecycle status.
PRD-FR-004	Timetable	Create class sessions, detect trainer/classroom/batch conflicts, publish changes.
PRD-FR-005	Hostel & logistics	Manage hostel rooms/beds, allocations, capacity and logistics/resources.
PRD-FR-006	Multilingual LMS	Provide courses, multimedia lessons, quizzes, progress and language selection.
PRD-FR-007	Offline learning	Cache learning content and persist learner activity locally; queue changes for synchronization.
PRD-FR-008	Attendance	Support face+liveness where supported, QR attendance, session checks and manual exception review.
PRD-FR-009	Assessments	Create assessments, record attempts, score and finalize outcomes.
PRD-FR-010	Certification	Issue digitally signed certificates and expose verification/revocation status.
PRD-FR-011	Skill passport	Map courses/assessments to structured skills and evidence.
PRD-FR-012	Career chatbot	Answer course/career/job questions using approved knowledge and bounded retrieval.
PRD-FR-013	Employer portal	Register/verify employers, create jobs, define skill requirements, review candidates.
PRD-FR-014	Candidate matching	Provide rule-based/hybrid matching and explain every recommendation.
PRD-FR-015	Application workflow	Track applications from applied through shortlisted, selected and joined.
PRD-FR-016	Outcome tracking	Capture employer-verified employment outcomes and retention/follow-up events.
PRD-FR-017	Analytics	Provide programme, institution, learning, certification and employment dashboards.
PRD-FR-018	Notifications	Deliver in-app and optional external

ID	Module	Requirement
		notifications for important events.
PRD-FR-019	Audit	Record sensitive changes, administrative approvals and credential lifecycle events.
PRD-FR-020	Integration	Expose versioned APIs and adapters for future NCCT/government systems.

8. Offline-First Product Behavior

The product is not “fully offline” in the sense of replicating the entire national ERP on every device. Offline capability is intentionally limited to frontline workflows that benefit from local availability: content access, learning progress, attendance capture, queued assessments/submissions and local operational reads/writes. National analytics and cross-center governance remain cloud-centric.

State	Expected product behavior
Online	Cloud authoritative data available; local cache refreshes and queued events synchronize.
Degraded	Continue reads from cache; queue writes; reduce media quality if required.
Offline	Continue local LMS, supported attendance, progress capture and queued submissions.
Reconnected	Detect pending events, validate integrity, deduplicate, reconcile conflicts, acknowledge.
Conflict	Do not silently overwrite critical fields; create review task for sensitive collisions.

Sync design basis

The research corpus recommends append-only change tracking, incremental synchronization and central reconciliation for hackathon-scale offline systems. CRDTs are reserved for future narrow collaboration use cases, not the entire relational ERP. [S2]

9. Attendance Product Requirements

Attendance is intentionally multimodal because the PS allows face recognition / QR and the research identifies spoofing, lighting, occlusion, liveness and privacy risks. [S1][S2]

Path	Behavior	Fallback
Face + liveness	Capture camera frame, quality check, liveness, match, confidence threshold; create attendance event only on policy-defined success.	QR / manual review
QR	Scan time-bounded/session-bound token and verify trainee/session.	Manual review

Path	Behavior	Fallback
Manual exception	Trainer/admin reviews unresolved attempts; resolution creates a signed/audited event.	N/A

9.1 Attendance acceptance rules

- A duplicate submission must be idempotent and not create double attendance.
- Attendance must retain the original event and any later correction as separate audit actions.
- Raw biometric media must not be uploaded by default; use data minimization and configurable retention.
- If a camera is unavailable, QR remains functional.
- If both biometric and QR are unavailable, manual exception remains possible for authorized staff.
- The system must not silently mark a failed recognition attempt as absence.

10. Credentials & Skill Passport

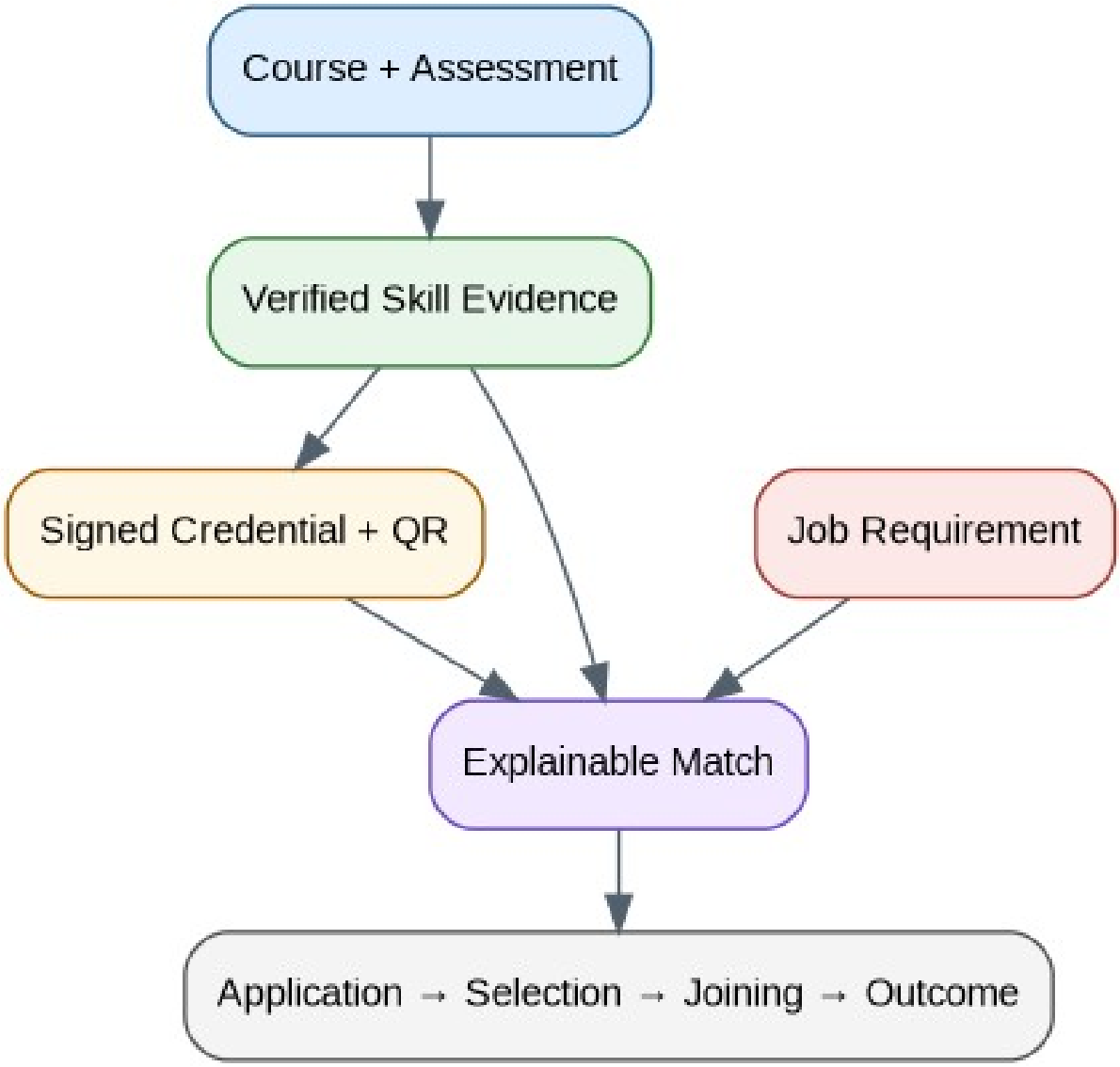


Figure 2. Evidence chain from learning to employment.

A certificate is treated as a verifiable evidence package rather than a static PDF. QR data references a credential record; the server verifies issuer signature and current status. Integration with national credential rails is an adapter target, not an MVP dependency. Research supplied with the project notes that DigiLocker already provides digital document rails and that signed verification with revocation is more defensible than a static QR-only certificate. [S2]

Credential field	Purpose
credential_id	Stable verification identifier
issuer_id	Institution/authority that issued it
subject_id	Trainee identifier
programme_id	Programme/batch reference

Credential field	Purpose
skills[]	Structured skill claims with evidence references
assessment_summary	High-level evidence without unnecessary sensitive data
issued_at / expires_at	Lifecycle timestamps where applicable
status	ACTIVE / REVOKED / SUSPENDED
signature	Cryptographic integrity proof

11. Employment & Outcomes

The employment module should be intentionally smaller than a commercial job marketplace. Its purpose is to connect verified cooperative skills to verified opportunities and expose the resulting outcome funnel.

Employer verification → Job posting → Skill requirements → Candidate discovery → Application → Selection → Joining → Employer verification → Outcome

11.1 Explainable matching

Matching should be deterministic or hybrid in the MVP. AI may parse a job description and explain the result, but the critical match should remain inspectable. The product should show which requirements are satisfied, what evidence supports each one, and which requirements are missing.

11.2 Employment verification

Status	Definition
REPORTED	Trainee reports an employment event.
EMPLOYER_CONFIRMED	Employer confirms the joining/outcome.
INSTITUTION_VERIFIED	Authorized institution validates according to its policy/evidence.
RETAINED	Follow-up indicates continued employment for a configured period.

12. AI & Multilingual Experience

AI is an assistive layer. The system must remain useful when AI is unavailable. The research corpus supports bounded/RAG-based assistants and notes that small offline RAG models are technically feasible, but clean degradation is more important for the MVP. [S2]

AI capability	MVP behavior	Offline behavior
Career Copilot	Retrieve approved programme/job knowledge and answer questions.	Cached FAQ/knowledge; no requirement for generative reasoning offline.
Learning Copilot	Explain lessons and answer course-	Cached explanations / local content

AI capability	MVP behavior	Offline behavior
	specific questions.	search.
Skill extraction	Convert course/assessment evidence into structured skill tags.	Precomputed skill mappings; full AI extraction when online.
Job matching explanation	Explain deterministic match results.	Existing cached job/candidate data only.
Admin intelligence	Flag anomalies and trends.	Precomputed local alerts only where implemented.

Bhashini is an integration option for multilingual services; its official site exposes APIs, AI models, language services, datasets and developer resources through organization registration. This PRD does not claim that Bhashini itself is fully offline. [W3]

13. Analytics & Monitoring

Domain	Metrics
Training	Enrolled, attendance, completion, dropout, batch utilization
Learning	Lesson completion, quiz scores, assessment performance, weak topics
Credentials	Issued, verified, revoked, verification requests
Employment	Applications, shortlist, selection, joining, employer confirmation, retention
Institution	Programme performance, trainer workload, hostel occupancy, resource utilization
Platform	Active users, sync lag, queued events, error rate, Local Fabric health

14. Security, Privacy & Trust

- Use RBAC and least privilege for all roles.
- Encrypt transport with TLS and encrypt sensitive local storage where supported.
- Store only the minimum data required for the specific workflow.
- Separate operational data, audit data and biometric-derived data logically and by access policy.
- Maintain explicit audit records for certificate issuance, revocation, attendance corrections, assessment finalization and employment verification.
- Do not assume Aadhaar integration.
- Provide consent/notice flows and retention/deletion controls for personal data according to the applicable legal regime and deployment contract.

Current legal design target

India has the Digital Personal Data Protection Act, 2023 and notified DPDP Rules, 2025 with phased commencement. The production implementation should revalidate which provisions are in force at deployment and obtain formal legal/NCCT review. The product design should nevertheless target clear notice, purpose limitation, data minimization, security and rights workflows from the start. [W1][W2]

15. Integration Strategy

NCCT-specific LMS/ERP/API details are not established in the research corpus. Therefore, integrations are represented as replaceable adapters rather than assumed direct dependencies. [S3]

Adapter	Purpose	MVP
NCCT Adapter	Future exchange with NCCT systems	Mock contract
Credential Adapter	Future DigiLocker/credential rail integration	Internal signed credential verifier
Bhashini Adapter	Translation / speech services	Optional online integration
Employer Adapter	External employer data or cooperative directories	Internal employer portal
Notification Adapter	SMS/email/WhatsApp or other approved channel	In-app first

16. Non-Functional Requirements

ID	Requirement
NFR-001 Availability	Core center workflows remain usable during temporary connectivity loss.
NFR-002 Offline durability	Committed local events survive application restart and power interruption to the extent supported by the host device/storage.
NFR-003 Sync integrity	Duplicate events are idempotent; malformed/untrusted events are quarantined.
NFR-004 Security	All privileged APIs enforce authentication + authorization; sensitive actions are audited.
NFR-005 Performance	Common interactive API requests target <500 ms server time under demo-scale load; sync is asynchronous.
NFR-006 Scalability	Cloud tier is stateless at API layer and can scale horizontally; database is authoritative.
NFR-007 Accessibility	Responsive/PWA interface works on common desktop/mobile dimensions and supports keyboard/touch.
NFR-008 Maintainability	Domain modules are independently testable and integrations are adapter-based.
NFR-009 Observability	Errors, sync lag, queue depth, latency and node health are

ID	Requirement
	measurable.
NFR-010 Recoverability	Backup/restore and local queue replay procedures are documented and testable.

17. MVP / Phase 2 / Future

MVP	Phase 2	Future
Programme + nominations	Advanced workflow automation	NCCT national orchestration
Profiles + batches	Advanced eligibility rules	Cross-ministry federation
Offline LMS + sync	More offline content types	Advanced edge fleet management
QR + face demo + manual review	Improved liveness/occlusion models	Production-grade biometric integrations
Assessments + signed QR credentials	External credential rail integration	Broader credential federation
Employer/job/application workflow	Skill taxonomy improvements	Large employer ecosystem
Career RAG chatbot	Multilingual voice	On-device advanced AI
Core analytics	Predictive analytics	National longitudinal policy analytics

18. PS Traceability

PS ID	Official requirement	Sahakar-Link coverage	Status
PS-01	Online programme registration / nominations	PRD-FR-001	MVP
PS-02	Participant/institution/trainee profiles	PRD-FR-002	MVP
PS-03	Interactive multilingual e-learning	PRD-FR-006	MVP
PS-04	Face recognition / QR attendance	PRD-FR-008	MVP
PS-05	Timetable, hostel, logistics	PRD-FR-004/005	MVP
PS-06	LMS + assessments + certification	PRD-FR-006/009/010	MVP
PS-07	Skill certification repository / verification	PRD-FR-011/010	MVP
PS-08	Career counseling chatbot	PRD-FR-012	MVP
PS-09	Employer/recruiter dashboard	PRD-FR-013/014/015	MVP
PS-10	Mobile + offline-accessible learning	PRD-FR-006/007	MVP
PS-11	Centralized database / monitoring	PRD-FR-017/020	MVP
PS-12	Cloud ERP / analytics / mobile /	PRD-FR-003..018	MVP

PS ID	Official requirement	Sahakar-Link coverage	Status
	multimedia		
PS-13	Software + Hardware mode	Hardware-agnostic Local Fabric + available devices; category interpretation must be confirmed	Risk / clarification

19. Success Metrics

Metric	Definition	Initial target for prototype
Offline continuity	Core LMS functions available with network disabled	100% of demo learning flow
Sync success	Queued valid events synchronized and acknowledged	>99% in controlled demo
Duplicate prevention	Duplicate event does not create duplicate state	100% on tested cases
Credential verification	Valid credential verifies; revoked credential fails	100% of demo cases
Attendance fallback	Failed face flow can complete via QR/manual review	100% of demo cases
Explainable matching	Every shown match includes requirement/evidence explanation	100% of displayed matches
Critical audit coverage	Sensitive actions create audit events	100% of defined critical actions

20. Risks, Assumptions & Open Questions

Item	Type	Treatment
NCCT current LMS/ERP not verified	Assumption	Adapter architecture; verify through official NCCT sources before production.
NCCT exact nomination workflow not verified	Assumption	Use configurable workflow model.
NCCT certificate format not verified	Assumption	Use normalized internal credential model; map external format later.
NCCT employer roster not verified	Assumption	Employer onboarding + verification; no invented roster.
Bhashini offline behavior not established	Assumption	Use Bhashini as optional online service; local fallback exists.
Hardware category interpretation	Risk	Confirm SIH category expectations with SPOC.
Biometric privacy / compliance	Risk	Use QR/manual fallback; data minimization; legal review before production.
Offline sync conflicts	Risk	Immutable events + versioned records +

Item	Type	Treatment
		manual review for critical fields.
Employer adoption	Risk	Keep MVP recruiter workflow small and measurable.

21. Acceptance Criteria

1. An admin can create a programme, open nominations and form a batch.
2. A trainee can access course content on a supported device while online.
3. The same trainee can continue core learning after network disconnection.
4. Progress/quiz/attendance events persist locally and survive reload/restart.
5. Queued events synchronize after reconnect without duplicate records.
6. Attendance can be completed by face/liveness demo, QR fallback or manual review.
7. Assessment completion can result in a signed credential with QR verification.
8. A recruiter can create a job and view explainable candidate matches.
9. A trainee can apply and the application can progress through defined outcome states.
10. NCCT/admin dashboards show training and employment funnel metrics.
11. Critical actions appear in the audit log.
12. The system remains functional in the absence of AI for core learning operations.

22. Release Plan

Sprint	Outcome
Sprint 1	Foundation: auth, roles, institutions, programmes, profiles, base schema
Sprint 2	Nomination, batches, timetable, hostel, learner dashboard
Sprint 3	LMS + local cache + SQLite/local node + offline state machine
Sprint 4	Attendance QR + face/liveness demo + exception queue
Sprint 5	Assessments + signed credentials + verification portal
Sprint 6	Employer, jobs, matching, applications, outcomes
Sprint 7	Career RAG + multilingual layer + analytics
Sprint 8	Hardening: sync conflict tests, security tests, demo rehearsal

23. Glossary

Term	Meaning
Local Fabric	Software services/cache/database running on available center hardware for local-first workflows.
Event Spine	Event queue + synchronization + reconciliation backbone.
Idempotency	Applying the same event twice produces the same final state as applying it once.

Term	Meaning
Skill Passport	Structured skill evidence linked to training and assessment.
Credential	Digitally signed proof of a training/certification outcome.
RAG	Retrieval-Augmented Generation using approved source knowledge.
Outcome	A post-training employment milestone or other configured programme result.

24. Evidence & References

S1 — Official SIH 2026 Problem Statement SIH26087 (provided in the project source) — https://sih.gov.in/sih2026PS

S2 — Offline-first / rural learning / edge research corpus supplied in the project workspace — Project research corpus; see internal source files supplied by the team

S3 — NCCT operational / government-platform research corpus supplied in the project workspace — Project research corpus; NCCT-specific details are explicitly marked unverified where evidence was absent

W1 — Ministry of Electronics and Information Technology — Digital Personal Data Protection Act, 2023 — https://www.meity.gov.in/static/uploads/2024/02/Digital-Personal-Data-Protection-Act-2023.pdf

W2 — MeitY — Digital Personal Data Protection Rules, 2025 — https://www.meity.gov.in/documents/act-and-policies/digital-personal-data-protection-rules-2025-gDOxUjMtQWa

W3 — Bhashini — National Language Translation Mission — https://bhashini.gov.in/

W4 — STQC — Biometric Devices Testing and Certification — https://www.stqc.gov.in/bio-metric-devices-testing-and-certification-0

W5 — DigiLocker — Government digital document platform (integration target; production access subject to onboarding/authorization) — https://www.digilocker.gov.in/

Selected research anchors from the supplied corpus include Hillier (2018) on offline e-learning; Dai et al. (2022) on offline collaborative learning; Linhalis et al. (2026) on offline LMS design; Maro et al. (2023/2024) on offline LMS deployment and micro-server feasibility; Surantha & Sugijakko (2024) and Abiodun et al. (2026) on lightweight face/liveness attendance; Tanaka (2026) and Stanelytė (2026) on verifiable vocational credentials; Wood et al. (2024) and Koca & Ortakaya (2023) on workplace training/employment transitions. These sources support architectural direction, not NCCT-specific operational facts.

Source discipline

NCCT-specific LMS/ERP, nomination, certificate, employer roster, infrastructure and API details remain unverified in the supplied research corpus. They are therefore represented as configurable design points or open questions, not facts.